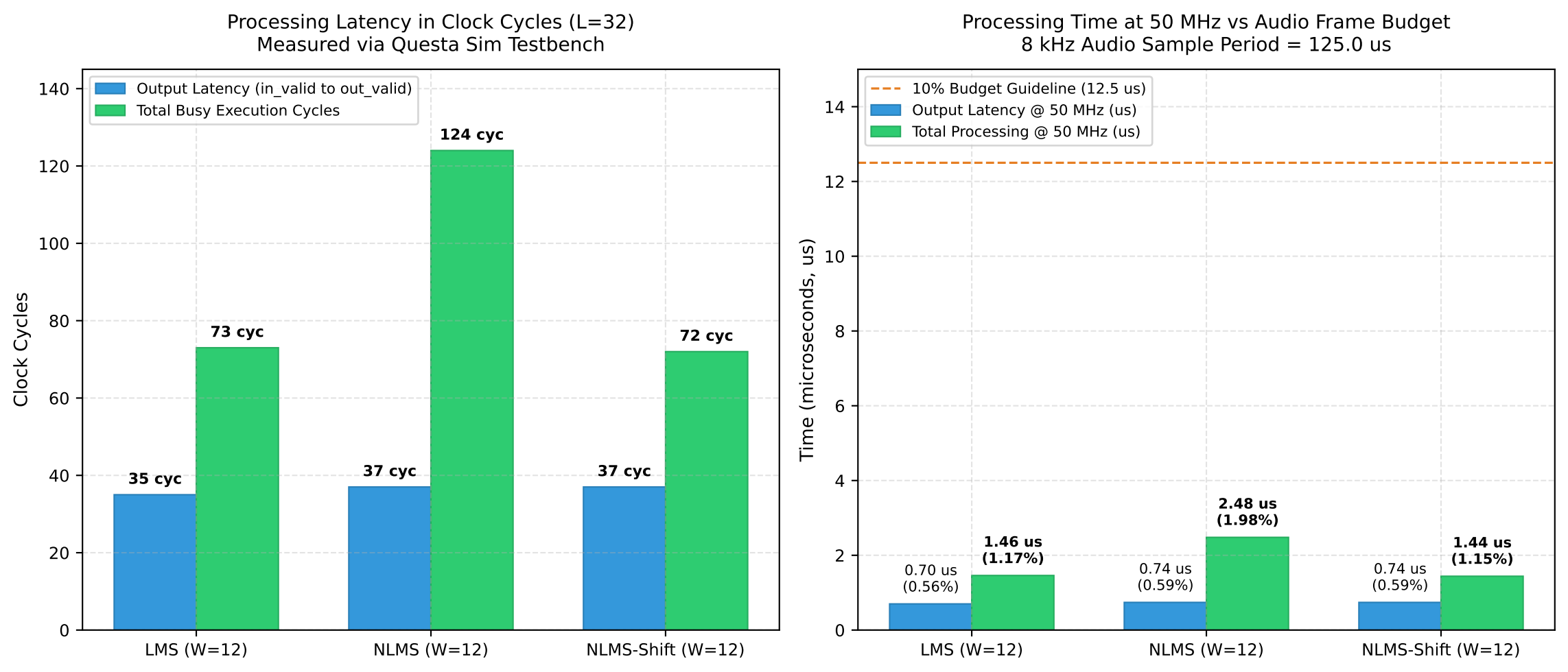
Processing Latency in Clock Cycles (L=32)
Measured via Questa Sim Testbench
Output Latency (in_valid to out_valid)
Total Busy Execution Cycles
Clock Cycles
0
20
40
60
80
100
120
140
35 cyc
73 cyc
37 cyc
124 cyc
37 cyc
72 cyc
LMS (W=12)
NLMS (W=12)
NLMS-Shift (W=12)
Processing Time at 50 MHz vs Audio Frame Budget
8 kHz Audio Sample Period = 125.0 us
10% Budget Guideline (12.5 us)
Output Latency @ 50 MHz (us)
Total Processing @ 50 MHz (us)
Time (microseconds, us)
0
2
4
6
8
10
12
14
0.70 us
(0.56%)
1.46 us
(1.17%)
0.74 us
(0.59%)
2.48 us
(1.98%)
0.74 us
(0.59%)
1.44 us
(1.15%)
LMS (W=12)
NLMS (W=12)
NLMS-Shift (W=12)